

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0209

Dranesville 138 kV
August 20, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0209. The Interconnection Customer proposes a 74.3 MW battery energy storage facility interconnecting at the Dranesville 138 kV substation in the Virginia area. The request was received on March 05, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 146 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,907,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Cub-Run Dp / Grand Blanc Generating Sta 138 KV line is overloaded to 133.6% of its 239 MVA emergency rating for the loss of the Yavapai / Luna 138 KV. The project contributes 5.3% DFAX (3.9 MW) to this overload.

For the loss of Shepard / Bloomington 138 KV Ckt 2, the Bloomington / Shepard 138 KV Ckt 1 loads to 132.9% of its 308 MVA rating. Project DFAX: 6.0%.

Jack McDonough / Low Gap 138 KV: 125.3% loading (180 MVA) following the outage of the Artesia Country Club / Clark County 345 KV line; DFAX = 6.0%.

For the loss of Shepard / Swinks Mill 138 KV Circuit 2, the Swinks Mill / Shepard Substation 138 KV #1 loads to 123.5% of its 350 MVA rating. Project DFAX: 6.0%.

For the loss of Bloomington / Mottman 138 KV Cir. 1, the Mottman / Bloomington 138 KV Ckt 2 loads to 121.8% of its 378 MVA rating. Project DFAX: 6.5%.

For the loss of Foster Dam / McCormick 138 KV, the Dranesville / Anthem 138 KV loads to 121.1% of its 242 MVA rating. Project DFAX: 6.3%.

The Shepard / Bloomington 138 KV is overloaded to 121.1% of its 337 MVA emergency rating for the loss of the Shepard / Bloomington 138 KV #1. The project contributes 6.0% DFAX (4.5 MW) to this overload.

For the loss of Carderock / Swinks Mill 138 KV Ckt 1, the Swinks Mill / Carderock 138 KV Cir. 2 loads to 120.5% of its 367 MVA rating. Project DFAX: 5.8%.

Carderock / Elk Garden 138 KV: 120.3% loading (239 MVA) following the outage of the Helijet / Elk Garden Substation 138 KV L; DFAX = 6.7%.

Swinks Mill / Shepard 138 KV Line 2: 116.2% loading (372 MVA) following the outage of the Swinks Mill / Shepard 138 KV Circuit 1; DFAX = 6.0%.

The Foster Dam / Elk Garden Substation 138 KV is overloaded to 110.2% of its 299 MVA emergency rating for the loss of the Dranesville / Anthem 138 KV L. The project contributes 7.0% DFAX (5.2 MW) to this overload.

Anthem / Elk Garden 138 KV Line: 109.9% loading (229 MVA) following the outage of the Foster Dam / McCormick 138 KV; DFAX = 6.3%.

For the loss of Mottman / Bloomington 138 KV Line, the Mottman / Bloomington 138 KV Line 1 loads to 109.8% of its 419 MVA rating. Project DFAX: 6.5%.

The Swinks Mill / Cardrock 138 KV #1 is overloaded to 109.5% of its 404 MVA emergency rating for the loss of the Swinks Mill / Carderock 138 KV Circuit 2. The project contributes 5.8% DFAX (4.3 MW) to this overload.

The Foster Dam / McCormick Substation 138 KV is overloaded to 107.6% of its 337 MVA emergency rating for the loss of the Anthem / Dranesville 138 KV. The project contributes 7.0% DFAX (5.2 MW) to this overload.

The Elk Garden / Helijet 138 KV L is overloaded to 106.1% of its 282 MVA emergency rating for the loss of the Elk Garden / Carderock 138 KV line. The project contributes 8.0% DFAX (6.0 MW) to this overload.

Colesville / Darien 138 KV: 103.1% loading (317 MVA) following the outage of the Adrian Energy Associates LLC 345/138 kV Transformer Bank; DFAX = 6.1%.

The Allegheny Energy Units 3 4 & 5 345/1 kV Transformer Bank is overloaded to 101.8% of its 951 MVA emergency rating for the loss of the Hilltop / Piney Crk Project 500 KV. The project contributes 5.6% DFAX (4.2 MW) to this overload.

Allegheny Energy Units 3 4 & 5 500 kV/1 kV XFMR: 101.6% loading (953 MVA) following the outage of the Hilltop / Piney Creek Project 500 KV; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cub-Run Dp / Grand Blanc Generating Sta 138 KV line	Rebuild line with higher-capacity conductor	\$3,680,000	25% (\$904,000)
Bloomington / Shepard 138 KV Ckt 1	Rebuild line with higher-capacity conductor	\$2,505,000	100% (\$2,505,000)
Jack McDonough / Low Gap 138 KV	Rebuild line with higher-capacity conductor	\$1,162,000	3% (\$32,000)
Swinks Mill / Shepard 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$1,340,000	100% (\$1,340,000)
Mottman / Bloomington 138 KV Ckt 2	Rebuild line with higher-capacity conductor	\$1,190,000	100% (\$1,190,000)
Dranesville / Anthem 138 KV	Rebuild line with higher-capacity conductor	\$3,621,000	100% (\$3,621,000)
Shepard / Bloomington 138 KV	Rebuild line with higher-capacity conductor	\$1,530,000	100% (\$1,530,000)
Swinks Mill / Carderock 138 KV	Rebuild line with higher-capacity conductor	\$3,294,000	100% (\$3,294,000)
Carderock / Elk Garden 138 KV	Rebuild line with higher-capacity conductor	\$2,191,000	8% (\$165,000)
Swinks Mil / Shepard 138 KV	Rebuild line with higher-capacity conductor	\$740,000	100% (\$740,000)
Foster Dam / Elk Garden Substation 138 KV	Rebuild line with higher-capacity conductor	\$8,656,000	6% (\$553,000)
Anthem / Elk Garden 138 KV Line	Reconductor line	\$2,846,000	100% (\$2,846,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bloomington / Mottman 138 KV Ckt 1	Reconductor line	\$431,000	100% (\$431,000)
Swinks Mill / Cardrock 138 KV #1	Reconductor line	\$1,294,000	100% (\$1,294,000)
Foster Dam / McCormick 138 KV	Reconductor line	\$345,000	6% (\$22,000)
Elk Garden / Helijet 138 KV	Reconductor line	\$374,000	8% (\$28,000)
Colesville / Darien 138 KV	Reconductor line	\$507,000	4% (\$20,000)
Allegheny Energy Units 3 4 & 5 345/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$19,323,000	1% (\$148,000)
Allegheny Energy Units 3 4 & 5 Substation 500 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$31,755,000	1% (\$244,000)
	Total allocated to project		\$20,907,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.